

Technical Report on Smartphone Camera System

Analysis of the Realme 11 5G Camera: Hardware, Image Processing Pipeline, and AI Integration

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1. Introduction

Smartphone cameras have evolved from simple imaging sensors into highly sophisticated computational photography systems powered by Artificial Intelligence (AI).

Modern devices combine advanced CMOS image sensors, dedicated Image Signal Processors (ISP), Neural Processing Units (NPU), and machine learning algorithms to produce high-quality photographs that often rival entry-level DSLR cameras.

This report analyzes the camera system of the **Realme 11 5G**, focusing on its hardware specifications, engineering implementation, raw image processing pipeline, stabilization mechanisms, and AI-driven computational photography features.

The objective is to understand how modern smartphones transform raw sensor data into visually enhanced images.

2. Technical Specifications

Primary Camera Specifications	
Feature	Specification
Phone Model	Realme 11 5G
Primary Sensor	Samsung ISOCELL HM6
Resolution	108 MP
Sensor Size	1/1.67 inch
Pixel Size	0.64 μm
Effective Pixel Size (after binning)	1.92 μm
Color Filter Array	9-in-1 Nona-Bayer
Aperture	f/1.75
Focal Length	24 mm (35mm equivalent)
Lens Elements	6P Lens
Autofocus	PDAF
OIS	No
EIS	Yes
Video Resolution	1080p @30fps
Flash	LED Flash

3. Camera Hardware Architecture

Sensor

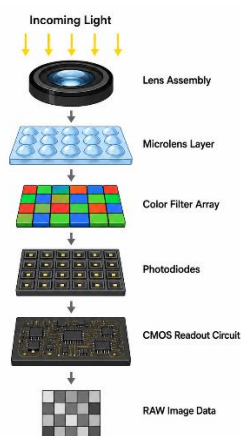
The Realme 11 5G uses the **Samsung ISOCELL HM6**, a 108 MP CMOS image sensor.

The sensor contains:

- Photodiodes
- Microlenses
- Color Filter Array
- Readout circuitry
- Analog amplifiers

The sensor converts incoming photons into electrical signals through the photoelectric effect.

Sensor Layout



Lens Assembly

The primary camera uses a 6-element plastic lens.

Advantages include:

- Better sharpness
- Lower distortion
- Reduced chromatic aberration

- Improved edge clarity

Aperture

The aperture is **f/1.75**.

A larger aperture allows:

- More light
- Better low-light performance
- Faster shutter speed
- Reduced image noise.

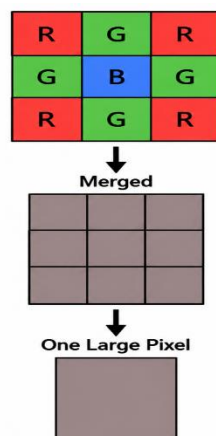
4. Pixel Architecture

Each pixel measures

0.64 μm

Very small pixels collect limited light individually.

To improve image quality, Samsung uses **Nona-Bayer technology**, grouping 9 neighboring pixels together.



Result:

Effective pixel size

1.92 μm

Benefits

- Better low-light images

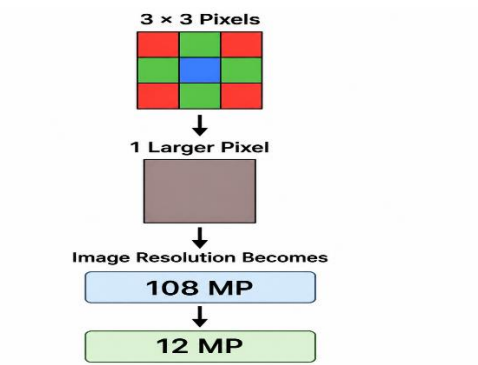
- Higher dynamic range
- Less image noise

5. Pixel Binning Technology

The Samsung HM6 sensor uses **9-in-1 Pixel Binning**.

Instead of outputting 108 MP every time,

the camera combines



Advantages

- More light captured
- Better night photography
- Less digital noise
- Faster image processing

6. Autofocus System

Uses

Phase Detection Autofocus (PDAF)

Working

Light enters through opposite sides of the lens.

Two images are compared.

If images mismatch



Lens motor moves.



Focus achieved.

Advantages

- Fast
- Accurate
- Continuous tracking

7. Artificial Intelligence Applications

- Modern smartphones use AI extensively.
- The Realme 11 5G integrates AI within the ISP and MediaTek Dimensity 6100+ chipset.
- AI identifies
 - Sky
 - Face
 - Hair
 - Grass
 - Buildings
 - Food
 - Water
 - Pets

Each region receives different processing.

Example

Sky



Increase blue saturation

Face



Smooth skin

Grass



Increase green saturation

8. Advantages

- 108 MP high-resolution sensor
- Excellent daylight photography
- Good low-light performance using pixel binning
- Fast PDAF autofocus
- AI-enhanced portrait mode
- HDR support
- Efficient ISP
- On-device AI processing

8.1. Limitations

- No Optical Image Stabilization
- Only 1080p video recording
- No ultrawide camera
- Small native pixel size
- Limited manual camera controls

9. Conclusion

The Realme 11 5G demonstrates how modern smartphones rely on computational photography rather than hardware alone to achieve high-quality images.

Although it lacks premium features such as Optical Image Stabilization (OIS) and 4K video recording, its Samsung ISOCELL HM6 108 MP sensor, 9-in-1 Nona-Bayer pixel binning, PDAF autofocus, and MediaTek Image Signal Processor enable excellent everyday photography.

The ISP performs essential operations including demosaicing, noise reduction, color correction, HDR merging, and image compression, while AI enhances portraits, night scenes, and semantic scene recognition.

This integration of advanced imaging hardware with on-device AI illustrates the evolution of smartphone cameras into intelligent imaging systems capable of producing detailed, vibrant, and well-balanced photographs under diverse conditions.

10. References

1. Samsung ISOCELL HM6 Sensor Datasheet.
2. Realme 11 5G Official Specifications.
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5. Gonzalez & Woods – *Digital Image Processing* (4th Edition).
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7. OpenCV Documentation.
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SENSORS

